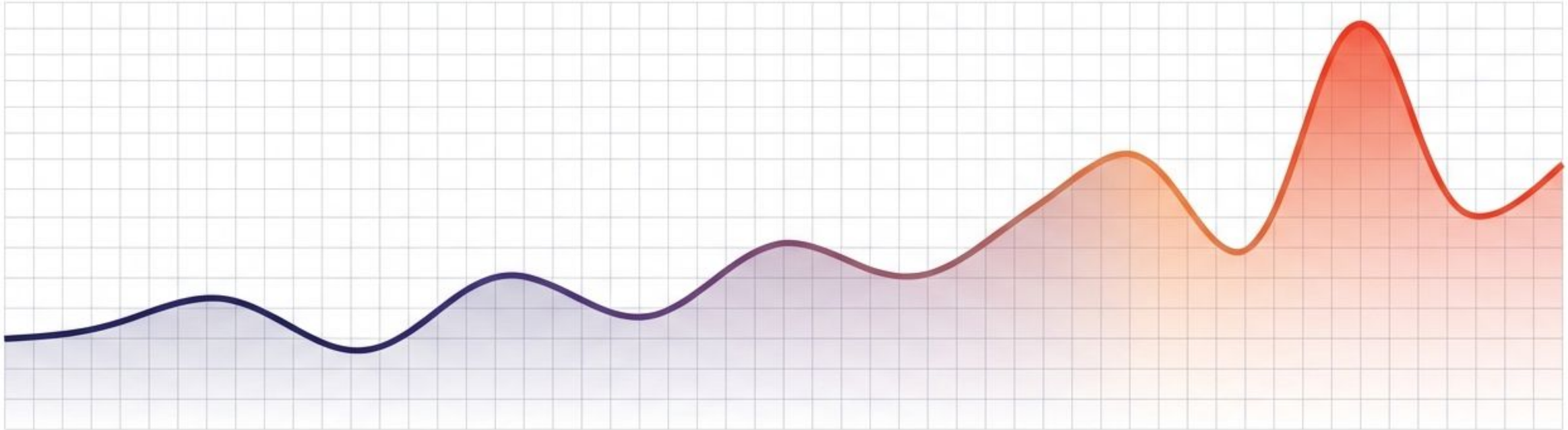
SYS_LOAD: 0.98
MEM_BW: 42GB/s
T_AMBIENT: 22C



MISD: Proactive Thermal Management for Edge AI Inference

Transitioning edge compute from reactive hardware sensors to predictive software proxies.

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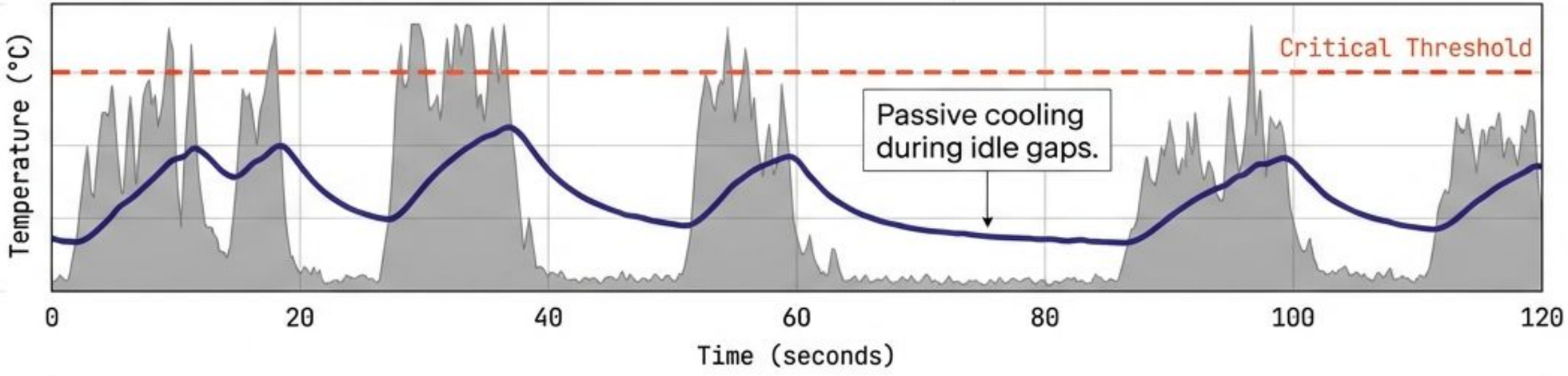
Small Language Models Break Legacy Legacy Thermal Governors

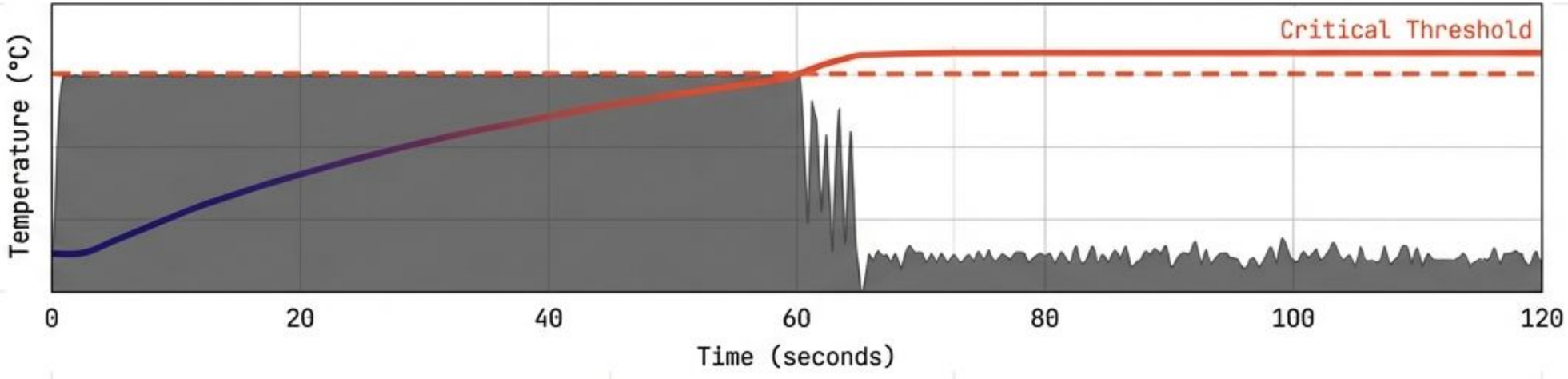
Bursty Workload (e.g., Web Browsing)



Standard Linux `thermal_core` relies on 10-100 ms polling intervals.

No idle gaps = no passive cooling.

SLM Inference



Result: Hard thermal throttling at 60 seconds.

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The Diagnostic Matrix of Hardware Monitoring Failures

	Indicator Type	Resolution	Blindspot
B0: Default Linux (step_wise)	Lagging	Silicon-level	Smoothed by die thermal mass
B1: DTS-Only Predictive	Lagging	Silicon-level	Smoothed by die thermal mass
B2: AMU-Only	Coarse	Utilization-level	Cannot distinguish math types
MISD (Proposed)	Leading ✓	Instruction-level ✓	None (Directly measures cause) ✓

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Isolating the Hidden Variable in Dynamic Power

$$P_{_dyn} = \alpha C V^2 f$$

α (Activity Factor)

The Hidden Variable. The fraction of transistors switching per cycle. Entirely dependent on the instruction stream and invisible to the OS.

C (Capacitance)

Hardware Fixed. Locked by the silicon physics.

V & f (Voltage & Frequency)

OS Controlled. The traditional levers of power management.

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Mathematical Instruction Set Density (MISD) as a Leading Indicator

$$\text{MISD} = \frac{(\text{ASE_SPEC} + \text{VFP_SPEC})}{\text{INST_RETIRED}}$$

MISD provides a software-level proxy for the silicon-level Activity Factor (α), tracking thermal density before heat physically materializes.

The MISD Predictive Advantage

t=0s

Instruction Execution: MISD detects high-density vector math.

t=3 to 5s

Thermal Mass Saturation: Silicon begins to heat up.

t=5+s

DTS Sensor Detection: Lagging indicator registers the heat.

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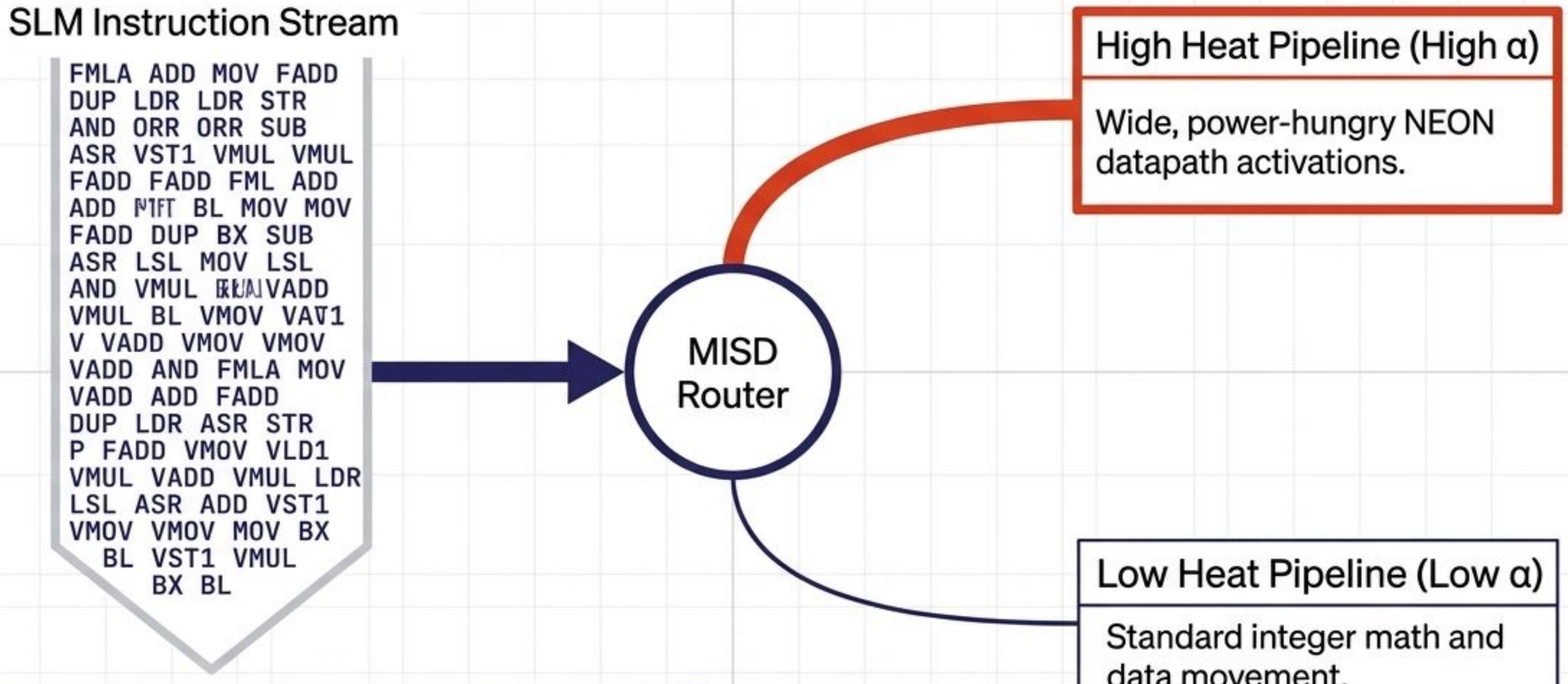
Aggregate Abstraction Maps SLM Thermal Trajectories



- We do not track individual instructions.
- Tracking the aggregate ratio of wide NEON activations versus basic integers provides a highly accurate, low-overhead predictor of thermal trajectory.

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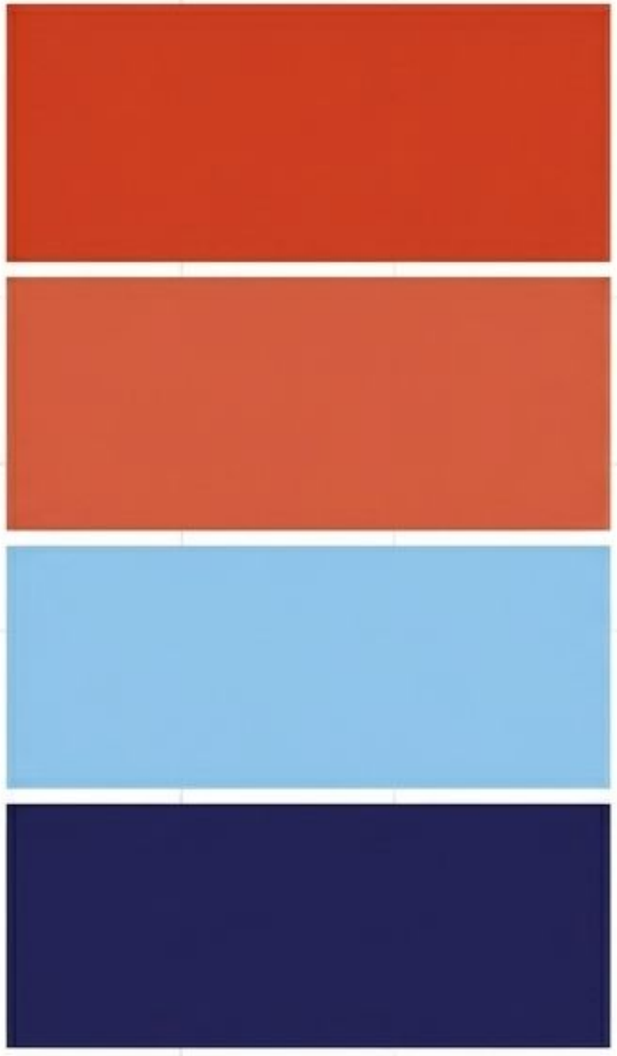
Empirical Calibration via Firestarter Kernels

Single-variable thermal stress tests lock C, V, f to isolate instruction mix as the sole driver of ΔT.

Mixed-mode variations (e.g., 80/20 FMLA to ADD) and programmed duty-cycle variations continuously train the model on transient SLM workload shifts.



Tier 1: HOT
`100% FMLA loop`
Max compute, highest α

Tier 2: WARM
`100% FADD loop`
Moderate α

Tier 3: COOL
`100% MOV & DUP loop`
Data movement

Tier 4: COLD
100% Integer ADD loop
Baseline scalar

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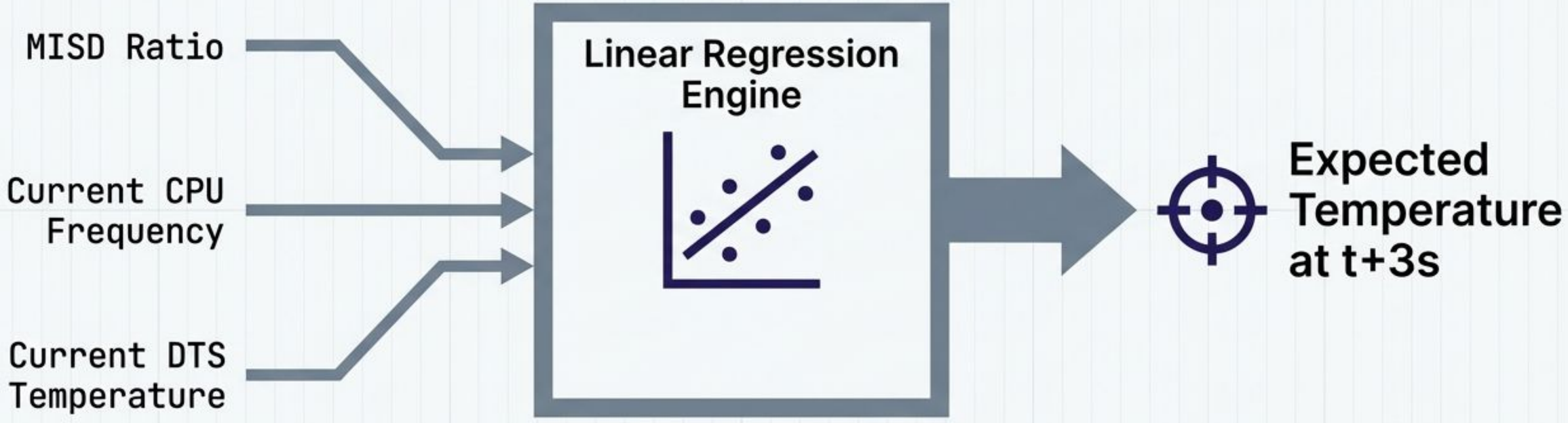
The Predictive Linear Regression Transfer Function



Translating instantaneous density metrics into a continuous, accurate transfer function mapping future thermal states.

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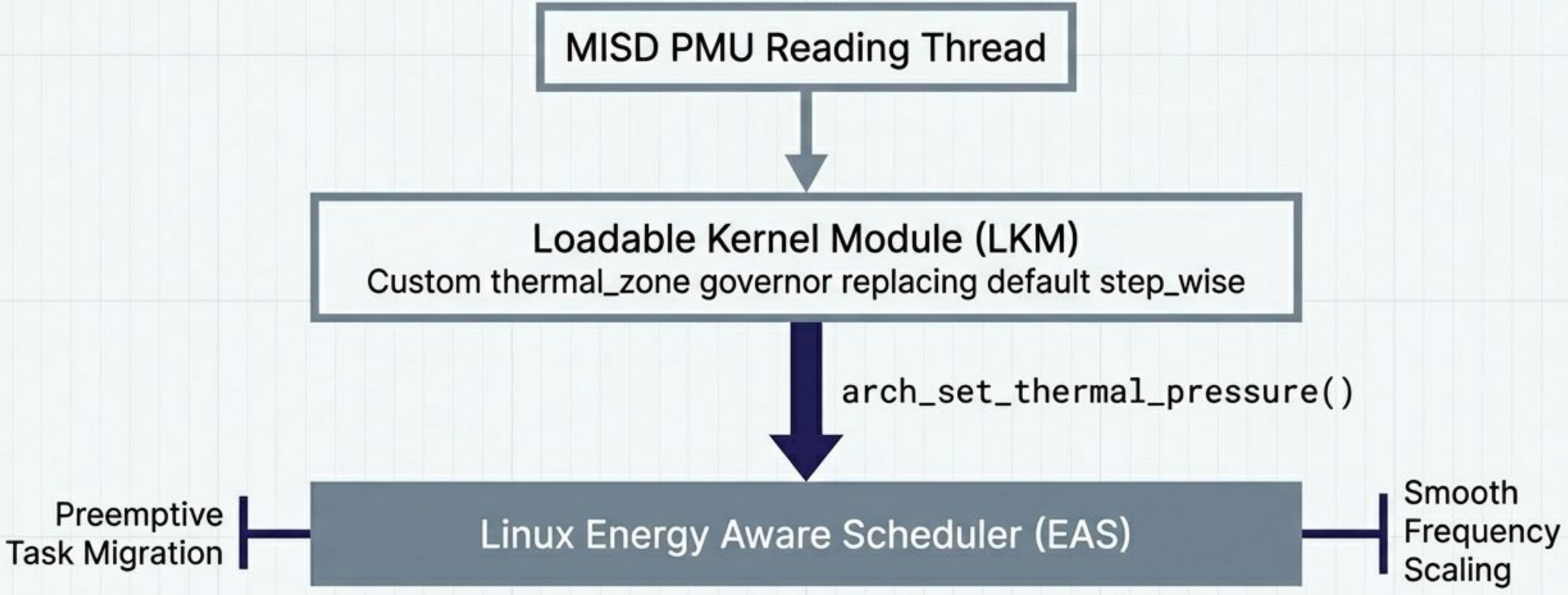
Autonomous Integration with the Linux Linux Energy Aware Scheduler



Instead of aggressive, direct frequency capping, MISD feeds ‘thermal pressure’ downward. EAS autonomously handles mitigation before hardware limits are breached.

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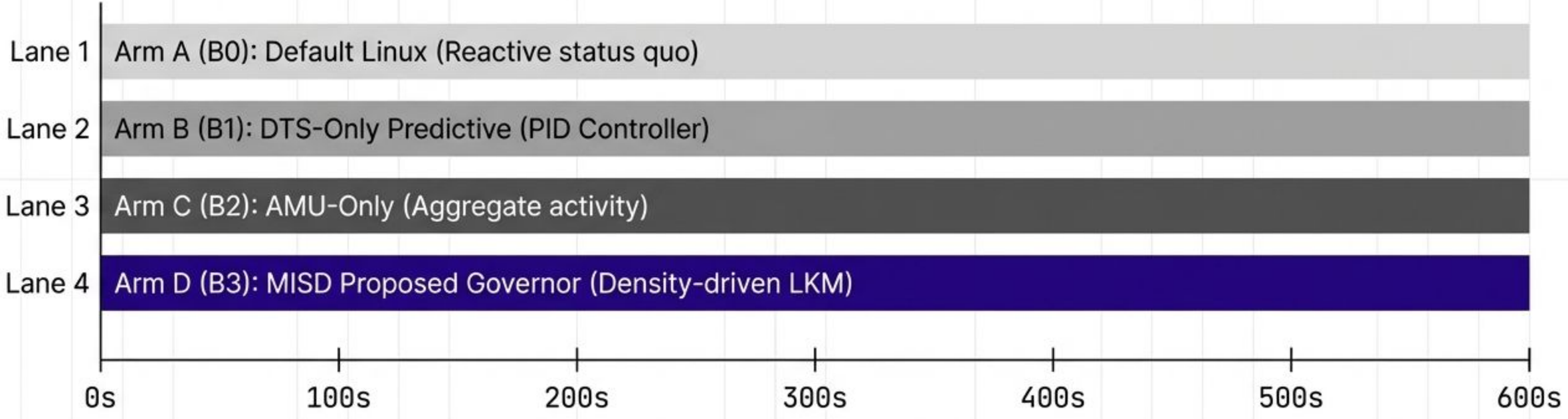
The Four-Arm Continuous Inference Experimental Design



Workload: Continuous SLM execution via llama.cpp.

Duration: 600 continuous seconds to ensure complete thermal saturation is achieved and maintained.

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Core Telemetry Metrics and Success Criteria

Average Tokens Per Second (TPS)

To prove sustained throughput improvements vs peak throughput.

Prediction Lead Time

Targeting a **3-5 second** predictive advantage between MISD indication and DTS confirmation.

False Positive Rate

Measuring the percentage of unnecessary throttling events.

Thermal Violation Count

Goal: Tracking **zero-tolerance** safety threshold breaches.

Sensor Overhead

Goal: Verifying the PMU reading thread consumes only **2-5%** of system resources.

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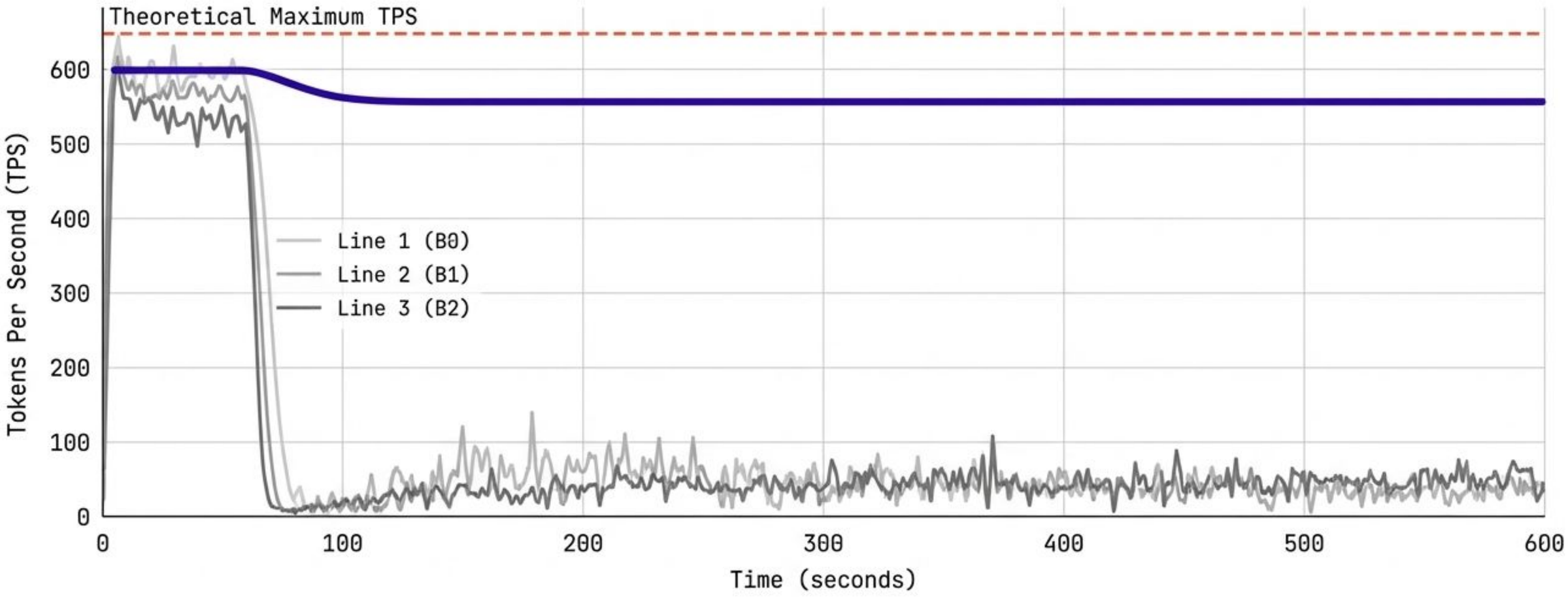
Throughput Stabilization Under Sustained Load



- Reactive models suffer catastrophic throughput collapse.
- MISD preemptively adjusts, locking in a sustained, stabilized performance plateau.

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Predictive Software Proxies Unlock Edge AI

Reactive Hardware Sensors

- Lagging indicators
- Unaware of instruction types
- Panic throttling
- Throughput collapse

Predictive Software Proxies

- Instruction-aware telemetry (MISD)
- 3-5 second predictive advantage
- Autonomous EAS task migration
- Stable, continuous throughput

Understanding what the core is executing is fundamentally superior to measuring how busy it is. MISD bridges the critical latency gap, providing the stable thermal foundation required for continuous Edge AI inference.

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